

Change Management: ERP Migration

Oracle to PostgreSQL migration -- impact analysis and rollback plan

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1. Introduction

This document describes the change management plan for migrating from Oracle Database 19c to the primary relational database as the primary database for billing and data warehousing workloads. The migration is driven by the constraint 'Legacy Oracle DB Migration Deadline Q4 2026' and the goal to Reduce IT Operational Costs by 25%.

For the broader cloud migration context, see the Cloud Migration Assessment (EA-CLD-009). The Vendor and Technology Assessment (EA-VND-014) provides the technical rationale for the PostgreSQL choice.

The incident management process defines four severity levels, with Severity 1 incidents requiring a response within fifteen minutes and a post-incident review within five business days. The migration strategy follows a strangler-fig pattern, where new functionality is built in the target platform while legacy features are gradually redirected; this approach minimizes disruption to active users and allows incremental validation of the new implementation. A proof-of-concept was conducted over a four-week period, involving representatives from the Claims Department, IT Operations, and the Data Analytics Team to validate the technical feasibility. A chaos engineering program was introduced in Q1 2025, starting with controlled fault injection in the staging environment to validate the resilience of the system against network partitions and dependency failures. The initiative is tracked in the enterprise PMO tool with bi-weekly status updates to the steering committee. Automated deployment pipelines leverage the CI/CD infrastructure described in the platform engineering documentation. The API design guidelines specify that all REST endpoints must support content negotiation, return standard HTTP status codes, include correlation identifiers in response headers, and provide machine-readable error bodies in RFC 7807 format. Documentation standards require that all architecture decisions are recorded as Architecture Decision Records using the ADR format proposed by Michael Nygard, stored alongside the codebase in the version control system. A comprehensive training program was developed for end users, including online modules, instructor-led sessions, and a sandbox environment for hands-on practice. Application health is assessed using a synthetic monitoring approach, where automated scripts simulate real user journeys every five minutes and report availability from three geographically distributed probe locations. The team uses a definition of done that includes code review by at least two peers, passing CI pipeline, updated architecture diagrams, and entry in the change log maintained by the Release Manager. Service-level indicators are collected continuously and aggregated into dashboards visible to both technical teams and business stakeholders. Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer. Risk mitigation strategies include regular checkpoint reviews, automated regression testing, and staged rollout procedures. The operational support model is defined in the Operational Runbook and includes on-call rotation and escalation procedures. In the context of insurance operations, peak load occurs during natural disaster events when claim submissions can spike by up to ten times the normal rate; the auto-scaling configuration accounts for this burst pattern. The architecture team maintains a dependency register that maps each component to its upstream providers and downstream consumers, updated quarterly. Interoperability with partner systems is achieved through the Partner Integration Hub, which exposes a curated subset of the internal API surface with additional rate limiting, schema validation, and audit logging applied at the gateway layer. The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints. Stakeholders across multiple departments have contributed requirements for this workstream. Scalability testing performed in the staging environment confirmed that the system handles a three-times-peak load without degradation, providing sufficient headroom for seasonal demand fluctuations.

Monitoring and alerting are integrated with the central observability stack based on Prometheus and Grafana. The ERP migration change management has been a critical area of focus for ArchiSurance since the digital transformation initiative began in 2023. Risk mitigation strategies include regular checkpoint reviews, automated regression testing, and staged rollout procedures. Interoperability with partner systems is achieved through the Partner Integration Hub, which exposes a curated subset of the internal API surface with additional rate limiting, schema validation, and audit logging applied at the gateway layer. The data classification exercise, completed in Q4 2024, assigned sensitivity labels to all data objects, informing encryption, access control, and retention policies. The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam. Collaboration with the vendor support team has been essential during the implementation phase; weekly technical syncs ensure that configuration changes are validated against the vendor's reference architecture before being applied to the production environment. The migration strategy follows a strangler-fig pattern, where new functionality is

built in the target platform while legacy features are gradually redirected; this approach minimizes disruption to active users and allows incremental validation of the new implementation. The incident management process defines four severity levels, with Severity 1 incidents requiring a response within fifteen minutes and a post-incident review within five business days.

In the context of insurance operations, peak load occurs during natural disaster events when claim submissions can spike by up to ten times the normal rate; the auto-scaling configuration accounts for this burst pattern. A chaos engineering program was introduced in Q1 2025, starting with controlled fault injection in the staging environment to validate the resilience of the system against network partitions and dependency failures. Capacity forecasting models project a twenty-five percent year-over-year increase in data volume, driven primarily by the digitization of paper-based processes and the expansion of the broker channel. Documentation standards require that all architecture decisions are recorded as Architecture Decision Records using the ADR format proposed by Michael Nygard, stored alongside the codebase in the version control system. Key performance indicators for the initiative include transaction throughput, error rate, mean time to detection, mean time to recovery, and user satisfaction scores collected through quarterly surveys. Change management activities include stakeholder communication, training sessions, and documentation updates. The API design guidelines specify that all REST endpoints must support content negotiation, return standard HTTP status codes, include correlation identifiers in response headers, and provide machine-readable error bodies in RFC 7807 format. The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints. The ERP migration change management has been a critical area of focus for ArchiSurance since the digital transformation initiative began in 2023.

Interoperability with partner systems is achieved through the Partner Integration Hub, which exposes a curated subset of the internal API surface with additional rate limiting, schema validation, and audit logging applied at the gateway layer. Capacity forecasting models project a twenty-five percent year-over-year increase in data volume, driven primarily by the digitization of paper-based processes and the expansion of the broker channel. Integration testing spans three environments: development, staging, and pre-production. A formal lessons-learned workshop is held at the end of each quarter, with outcomes published to the internal wiki and action items assigned to specific team members with target completion dates. ArchiSurance participates in an industry-wide information sharing and analysis center for insurance companies, contributing anonymized threat intelligence data and receiving early warnings about sector-specific attack patterns. The solution leverages event-driven patterns wherever possible, with Apache Kafka providing durable, ordered event streams that decouple producers from consumers and enable replay for debugging and reprocessing scenarios. Collaboration with the vendor support team has been essential during the implementation phase; weekly technical syncs ensure that configuration changes are validated against the vendor's reference architecture before being applied to the production environment. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days. Key performance indicators for the initiative include transaction throughput, error rate, mean time to detection, mean time to recovery, and user satisfaction scores collected through quarterly surveys.

2. Migration Scope

The migration scope includes the following systems currently on Oracle: Billing System (accessing Commission Ledger), Data Warehouse (storing Analytics Cube data). Downstream consumers include the Batch Processing Engine and the Analytics Dashboard.

The enterprise data model, maintained by the Data Architecture team, defines canonical schemas for all shared data objects; integrating systems are expected to conform to these schemas or implement transformation layers that map proprietary formats to the canonical model. Performance benchmarks are captured in the enterprise monitoring framework and reported on a monthly basis to leadership. The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards. A dedicated Slack channel was established for real-time coordination between development, operations, and security teams during the rollout period. Capacity planning exercises revealed that the current infrastructure can sustain projected growth for the next twelve months, after which additional compute and storage resources will be required. A proof-of-concept was conducted over a four-week period, involving representatives from the Claims Department, IT Operations, and the Data Analytics Team to validate the

technical feasibility. Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer. The design considers both current-state constraints and target-state aspirations as documented in the strategic roadmap. Interoperability with partner systems is achieved through the Partner Integration Hub, which exposes a curated subset of the internal API surface with additional rate limiting, schema validation, and audit logging applied at the gateway layer. The transition plan was developed in alignment with the enterprise change management framework, incorporating staged rollouts, parallel-run periods, and defined go/no-go criteria at each milestone. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days. The business process documentation follows the BPMN 2.0 standard, with process models maintained in the Signavio process modeling tool and linked to the ArchiMate model elements in Sparx EA. The initiative is tracked in the enterprise PMO tool with bi-weekly status updates to the steering committee. The solution leverages event-driven patterns wherever possible, with Apache Kafka providing durable, ordered event streams that decouple producers from consumers and enable replay for debugging and reprocessing scenarios. A chaos engineering program was introduced in Q1 2025, starting with controlled fault injection in the staging environment to validate the resilience of the system against network partitions and dependency failures. The incident management process defines four severity levels, with Severity 1 incidents requiring a response within fifteen minutes and a post-incident review within five business days. The team uses a definition of done that includes code review by at least two peers, passing CI pipeline, updated architecture diagrams, and entry in the change log maintained by the Release Manager. Feedback from the user acceptance testing phase indicated that the new interface reduced the average task completion time by approximately thirty percent compared to the legacy system, though some users requested additional keyboard shortcuts for power-user workflows. External consultants from Deloitte provided an independent assessment of the architecture in late 2024. Change management activities include stakeholder communication, training sessions, and documentation updates. The architecture decisions were reviewed and approved by the Architecture Review Board in Q2 2024. Capacity forecasting models project a twenty-five percent year-over-year increase in data volume, driven primarily by the digitization of paper-based processes and the expansion of the broker channel. Stakeholders across multiple departments have contributed requirements for this workstream. Compliance with the General Data Protection Regulation requires that personally identifiable information is encrypted at rest and in transit, with access restricted to authorized roles as defined in the Identity and Access Management policy. The configuration management database maintains a record of all IT assets, including hardware serial numbers, software license keys, patch levels, and relationships between configuration items. Integration testing spans three environments: development, staging, and pre-production.

Interoperability with partner systems is achieved through the Partner Integration Hub, which exposes a curated subset of the internal API surface with additional rate limiting, schema validation, and audit logging applied at the gateway layer. Change management activities include stakeholder communication, training sessions, and documentation updates. The API design guidelines specify that all REST endpoints must support content negotiation, return standard HTTP status codes, include correlation identifiers in response headers, and provide machine-readable error bodies in RFC 7807 format. Regression test suites are maintained by the QA team and executed automatically on every merge to the main branch via the CI/CD pipeline. The architecture review confirmed that the proposed solution aligns with the TOGAF Architecture Development Method phases used by ArchiSurance, specifically phases B through D covering Business, Information Systems, and Technology Architecture. ArchiSurance participates in an industry-wide information sharing and analysis center for insurance companies, contributing anonymized threat intelligence data and receiving early warnings about sector-specific attack patterns. Scalability testing performed in the staging environment confirmed that the system handles a three-times-peak load without degradation, providing sufficient headroom for seasonal demand fluctuations. The architecture decisions were reviewed and approved by the Architecture Review Board in Q2 2024. Risk mitigation strategies include regular checkpoint reviews, automated regression testing, and staged rollout procedures.

The enterprise data model, maintained by the Data Architecture team, defines canonical schemas for all shared data objects; integrating systems are expected to conform to these schemas or implement transformation layers that map proprietary formats to the canonical model. Budget allocation was approved as part of the annual IT investment portfolio review conducted in November 2024. The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam. The design considers both current-state constraints and target-state aspirations as documented in the strategic roadmap. Key performance indicators for the initiative include transaction throughput, error rate, mean time to detection, mean time to recovery, and user satisfaction scores collected through quarterly surveys. Integration testing spans three environments: development,

staging, and pre-production. A comprehensive training program was developed for end users, including online modules, instructor-led sessions, and a sandbox environment for hands-on practice. The data retention policy mandates that transactional records are kept for seven years in online storage and archived to cold storage thereafter, with the ability to retrieve archived records within twenty-four hours upon regulatory request. The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards.

ArchiSurance participates in an industry-wide information sharing and analysis center for insurance companies, contributing anonymized threat intelligence data and receiving early warnings about sector-specific attack patterns. Monitoring and alerting are integrated with the central observability stack based on Prometheus and Grafana. The enterprise identity provider federates with the national eHerkenning system for business-to-government interactions and with DigiD for customer-facing authentication on the self-service portal. Dependencies on third-party services were documented and risk-assessed, with fallback mechanisms defined for each critical integration point. The operational support model is defined in the Operational Runbook and includes on-call rotation and escalation procedures. Risk mitigation strategies include regular checkpoint reviews, automated regression testing, and staged rollout procedures. Automated deployment pipelines leverage the CI/CD infrastructure described in the platform engineering documentation. A chaos engineering program was introduced in Q1 2025, starting with controlled fault injection in the staging environment to validate the resilience of the system against network partitions and dependency failures. The architecture decisions were reviewed and approved by the Architecture Review Board in Q2 2024.

3. Impact Analysis

The migration will impact the following business processes: Broker Commission Settlement (directly depends on Billing), Premium Calculation (through billing integration), and Regulatory Reporting (through the Data Warehouse). The Finance Department is the primary business stakeholder.

Integration impact: The ESB routes messages to the billing system. Connection strings and message schemas may need updates. The Batch Processing Engine ETL pipelines require Oracle-to-PostgreSQL SQL dialect conversion.

Data privacy impact assessment was filed with the Data Protection Officer in Q1 2025. Stakeholders across multiple departments have contributed requirements for this workstream. A proof-of-concept was conducted over a four-week period, involving representatives from the Claims Department, IT Operations, and the Data Analytics Team to validate the technical feasibility. Cross-functional workshops held in January and March 2025 helped refine the scope and integration points. Regression test suites are maintained by the QA team and executed automatically on every merge to the main branch via the CI/CD pipeline. External consultants from Deloitte provided an independent assessment of the architecture in late 2024. The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards. The Architecture Review Board convened a special session in February 2025 to evaluate the implications of the proposed changes on downstream systems and business processes. Change management activities include stakeholder communication, training sessions, and documentation updates. The initiative is tracked in the enterprise PMO tool with bi-weekly status updates to the steering committee. Security review was completed in accordance with the CISO's security-by-design framework. The architecture review confirmed that the proposed solution aligns with the TOGAF Architecture Development Method phases used by ArchiSurance, specifically phases B through D covering Business, Information Systems, and Technology Architecture. The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints. The business process documentation follows the BPMN 2.0 standard, with process models maintained in the Signavio process modeling tool and linked to the ArchiMate model elements in Sparx EA. Capacity planning exercises revealed that the current infrastructure can sustain projected growth for the next twelve months, after which additional compute and storage resources will be required. The non-functional requirements specify a maximum end-to-end latency of five hundred milliseconds at the ninety-ninth percentile, measured under a sustained load of one thousand concurrent users. The enterprise data model, maintained by the Data Architecture team, defines canonical schemas for all shared data objects; integrating systems are expected to conform to these schemas or implement transformation layers that map proprietary formats to the canonical model. Key performance indicators for the initiative include transaction throughput, error rate, mean time to detection, mean time to recovery, and user satisfaction scores collected

through quarterly surveys. Technical debt in this area was cataloged during a dedicated sprint in Q3 2024; the resulting backlog of forty-seven items has been prioritized by business impact and is being addressed alongside feature development. Feedback from the user acceptance testing phase indicated that the new interface reduced the average task completion time by approximately thirty percent compared to the legacy system, though some users requested additional keyboard shortcuts for power-user workflows. ArchiSurance participates in an industry-wide information sharing and analysis center for insurance companies, contributing anonymized threat intelligence data and receiving early warnings about sector-specific attack patterns. In the context of insurance operations, peak load occurs during natural disaster events when claim submissions can spike by up to ten times the normal rate; the auto-scaling configuration accounts for this burst pattern. During the initial assessment phase, the team identified several areas where the existing architecture fell short of the target-state requirements defined in the enterprise roadmap. Budget allocation was approved as part of the annual IT investment portfolio review conducted in November 2024. An accessibility audit was conducted by an external specialist in January 2025 to ensure that all user-facing components meet WCAG 2.1 Level AA standards, with remediation items tracked in the product backlog. The platform engineering team provides a standardized deployment template that all application teams are required to use, ensuring consistency in logging, health checks, and resource limits. The data classification exercise, completed in Q4 2024, assigned sensitivity labels to all data objects, informing encryption, access control, and retention policies. The data retention policy mandates that transactional records are kept for seven years in online storage and archived to cold storage thereafter, with the ability to retrieve archived records within twenty-four hours upon regulatory request. Scalability testing performed in the staging environment confirmed that the system handles a three-times-peak load without degradation, providing sufficient headroom for seasonal demand fluctuations.

The design considers both current-state constraints and target-state aspirations as documented in the strategic roadmap. Application health is assessed using a synthetic monitoring approach, where automated scripts simulate real user journeys every five minutes and report availability from three geographically distributed probe locations. A proof-of-concept was conducted over a four-week period, involving representatives from the Claims Department, IT Operations, and the Data Analytics Team to validate the technical feasibility. Collaboration with the vendor support team has been essential during the implementation phase; weekly technical syncs ensure that configuration changes are validated against the vendor's reference architecture before being applied to the production environment. Change management activities include stakeholder communication, training sessions, and documentation updates. Integration testing spans three environments: development, staging, and pre-production. An internal audit in March 2025 confirmed that the controls implemented for this area meet the requirements of the enterprise risk management framework. Automated deployment pipelines leverage the CI/CD infrastructure described in the platform engineering documentation.

Technical debt in this area was cataloged during a dedicated sprint in Q3 2024; the resulting backlog of forty-seven items has been prioritized by business impact and is being addressed alongside feature development. The Architecture Review Board convened a special session in February 2025 to evaluate the implications of the proposed changes on downstream systems and business processes. Feedback from the user acceptance testing phase indicated that the new interface reduced the average task completion time by approximately thirty percent compared to the legacy system, though some users requested additional keyboard shortcuts for power-user workflows. Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer. The incident management process defines four severity levels, with Severity 1 incidents requiring a response within fifteen minutes and a post-incident review within five business days. The team uses a definition of done that includes code review by at least two peers, passing CI pipeline, updated architecture diagrams, and entry in the change log maintained by the Release Manager. Post-implementation review sessions are scheduled for thirty and ninety days after go-live to capture lessons learned and identify optimization opportunities. The design considers both current-state constraints and target-state aspirations as documented in the strategic roadmap.

The vendor selection process followed the procurement guidelines established by the CFO's office. External consultants from Deloitte provided an independent assessment of the architecture in late 2024. The architecture team conducted a threat modeling exercise using the STRIDE framework, identifying potential spoofing, tampering, and information disclosure risks; mitigations for each threat were documented and tracked in the security risk register. During the initial assessment phase, the team identified several areas where the existing architecture fell short of the target-state requirements defined in the enterprise roadmap. The architecture decisions were reviewed and approved by the Architecture Review Board in Q2 2024. A chaos engineering program was introduced in Q1 2025, starting with

controlled fault injection in the staging environment to validate the resilience of the system against network partitions and dependency failures. Performance benchmarks are captured in the enterprise monitoring framework and reported on a monthly basis to leadership. The impact analysis business process has been a critical area of focus for ArchiSurance since the digital transformation initiative began in 2023.

4. Data Migration Plan

Data will be migrated from Oracle on the Database Server Primary to PostgreSQL on the same hardware (in-place migration). The Database Server Replica will serve as the PostgreSQL standby. Full backups will be stored on the SAN Storage Array before migration begins.

Migration SQL Example

```
-- Oracle to PostgreSQL migration: Commission Ledger
-- Step 1: Export from Oracle
expdp system/** schemas=BILLING dumpfile=billing_export.dmp

-- Step 2: Convert schema using ora2pg
ora2pg -t TABLE -o billing_tables.sql -s BILLING
ora2pg -t INSERT -o billing_data.sql -s BILLING

-- Step 3: Import into PostgreSQL
psql -h ams-db-01 -U archisurance -d billing -f billing_tables.sql
psql -h ams-db-01 -U archisurance -d billing -f billing_data.sql

-- Step 4: Verify row counts
SELECT 'policies' AS tbl, count(*) FROM policies
UNION ALL
SELECT 'commissions', count(*) FROM commissions
UNION ALL
SELECT 'invoices', count(*) FROM invoices;
```

A proof-of-concept was conducted over a four-week period, involving representatives from the Claims Department, IT Operations, and the Data Analytics Team to validate the technical feasibility. Lessons learned from the pilot phase have been incorporated into the current design iteration. Application health is assessed using a synthetic monitoring approach, where automated scripts simulate real user journeys every five minutes and report availability from three geographically distributed probe locations. The platform engineering team provides a standardized deployment template that all application teams are required to use, ensuring consistency in logging, health checks, and resource limits. Technical debt in this area was cataloged during a dedicated sprint in Q3 2024; the resulting backlog of forty-seven items has been prioritized by business impact and is being addressed alongside feature development. The information security team requires that all new systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval. Data privacy impact assessment was filed with the Data Protection Officer in Q1 2025. The solution was designed to be modular, allowing individual components to be upgraded or replaced without affecting the overall system availability or data integrity. Automated deployment pipelines leverage the CI/CD infrastructure described in the platform engineering documentation. The data classification exercise, completed in Q4 2024, assigned sensitivity labels to all data objects, informing encryption, access control, and retention policies. The data retention policy mandates that transactional records are kept for seven years in online storage and archived to cold storage thereafter, with the ability to retrieve archived records within twenty-four hours upon regulatory request. Scalability testing performed in the staging environment confirmed that the system handles a three-times-peak load without degradation, providing sufficient headroom for seasonal demand fluctuations. The enterprise technology radar, published semi-annually by the CTO's office, classifies technologies as Adopt, Trial, Assess, or Hold; the technologies used in this area are all currently in the Adopt or Trial rings. The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner. The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam. Risk mitigation strategies include regular checkpoint reviews, automated regression testing, and staged rollout procedures. The total cost of ownership analysis, prepared by the Finance Department in collaboration with IT, indicated a payback period of

approximately eighteen months assuming current utilization rates. The solution leverages event-driven patterns wherever possible, with Apache Kafka providing durable, ordered event streams that decouple producers from consumers and enable replay for debugging and reprocessing scenarios.

The incident management process defines four severity levels, with Severity 1 incidents requiring a response within fifteen minutes and a post-incident review within five business days. Lessons learned from the pilot phase have been incorporated into the current design iteration. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days. Business continuity considerations were incorporated from the outset, with the DR Recovery Node in the secondary data center providing failover capacity; the most recent failover test was conducted in October 2024 and achieved recovery within the target RTO of one hour. Technical debt in this area was cataloged during a dedicated sprint in Q3 2024; the resulting backlog of forty-seven items has been prioritized by business impact and is being addressed alongside feature development. The data migration planning Oracle to PostgreSQL has been a critical area of focus for ArchiSurance since the digital transformation initiative began in 2023. The business process documentation follows the BPMN 2.0 standard, with process models maintained in the Signavio process modeling tool and linked to the ArchiMate model elements in Sparx EA. The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner. The transition plan was developed in alignment with the enterprise change management framework, incorporating staged rollouts, parallel-run periods, and defined go/no-go criteria at each milestone.

Integration testing spans three environments: development, staging, and pre-production. The network team conducted a bandwidth utilization study in February 2025, confirming that inter-datacenter traffic remains below sixty percent of the available WAN capacity under normal operating conditions. External consultants from Deloitte provided an independent assessment of the architecture in late 2024. Compliance with the General Data Protection Regulation requires that personally identifiable information is encrypted at rest and in transit, with access restricted to authorized roles as defined in the Identity and Access Management policy. The API design guidelines specify that all REST endpoints must support content negotiation, return standard HTTP status codes, include correlation identifiers in response headers, and provide machine-readable error bodies in RFC 7807 format. The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline. The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints. An accessibility audit was conducted by an external specialist in January 2025 to ensure that all user-facing components meet WCAG 2.1 Level AA standards, with remediation items tracked in the product backlog. The solution was designed to be modular, allowing individual components to be upgraded or replaced without affecting the overall system availability or data integrity.

An internal audit in March 2025 confirmed that the controls implemented for this area meet the requirements of the enterprise risk management framework. Cross-functional workshops held in January and March 2025 helped refine the scope and integration points. The transition plan was developed in alignment with the enterprise change management framework, incorporating staged rollouts, parallel-run periods, and defined go/no-go criteria at each milestone. Security review was completed in accordance with the CISO's security-by-design framework. Monitoring and alerting are integrated with the central observability stack based on Prometheus and Grafana. The information security team requires that all new systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval. The architecture review confirmed that the proposed solution aligns with the TOGAF Architecture Development Method phases used by ArchiSurance, specifically phases B through D covering Business, Information Systems, and Technology Architecture. A proof-of-concept was conducted over a four-week period, involving representatives from the Claims Department, IT Operations, and the Data Analytics Team to validate the technical feasibility. During the initial assessment phase, the team identified several areas where the existing architecture fell short of the target-state requirements defined in the enterprise roadmap.

5. Application Changes

The Billing System will require connection string changes and Oracle-specific SQL conversion (e.g., ROWNUM to LIMIT/OFFSET, NVL to COALESCE). The Data Warehouse ETL stored procedures must be rewritten. The Analytics Dashboard remains on Microsoft SQL Server 2022 and is unaffected. The Customer Management Platform and PAS

already use PostgreSQL and are unaffected.

The architecture team maintains a dependency register that maps each component to its upstream providers and downstream consumers, updated quarterly. Collaboration with the vendor support team has been essential during the implementation phase; weekly technical syncs ensure that configuration changes are validated against the vendor's reference architecture before being applied to the production environment. The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards. The design considers both current-state constraints and target-state aspirations as documented in the strategic roadmap. The initiative is tracked in the enterprise PMO tool with bi-weekly status updates to the steering committee. Cross-functional workshops held in January and March 2025 helped refine the scope and integration points. ArchiSurance participates in an industry-wide information sharing and analysis center for insurance companies, contributing anonymized threat intelligence data and receiving early warnings about sector-specific attack patterns. A proof-of-concept was conducted over a four-week period, involving representatives from the Claims Department, IT Operations, and the Data Analytics Team to validate the technical feasibility. During the initial assessment phase, the team identified several areas where the existing architecture fell short of the target-state requirements defined in the enterprise roadmap. Implementation follows an iterative approach, with quarterly milestones aligned to the overall program timeline. A chaos engineering program was introduced in Q1 2025, starting with controlled fault injection in the staging environment to validate the resilience of the system against network partitions and dependency failures. In the context of insurance operations, peak load occurs during natural disaster events when claim submissions can spike by up to ten times the normal rate; the auto-scaling configuration accounts for this burst pattern. Security review was completed in accordance with the CISO's security-by-design framework. The operational support model is defined in the Operational Runbook and includes on-call rotation and escalation procedures. The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner. The team uses a definition of done that includes code review by at least two peers, passing CI pipeline, updated architecture diagrams, and entry in the change log maintained by the Release Manager. The architectural principles that guide this work include separation of concerns, loose coupling between components, preference for open standards, and the ability to scale horizontally under peak load conditions such as end-of-year policy renewals. The solution was designed to be modular, allowing individual components to be upgraded or replaced without affecting the overall system availability or data integrity. External consultants from Deloitte provided an independent assessment of the architecture in late 2024. Feedback from the user acceptance testing phase indicated that the new interface reduced the average task completion time by approximately thirty percent compared to the legacy system, though some users requested additional keyboard shortcuts for power-user workflows. The enterprise data model, maintained by the Data Architecture team, defines canonical schemas for all shared data objects; integrating systems are expected to conform to these schemas or implement transformation layers that map proprietary formats to the canonical model. Scalability testing performed in the staging environment confirmed that the system handles a three-times-peak load without degradation, providing sufficient headroom for seasonal demand fluctuations. The business process documentation follows the BPMN 2.0 standard, with process models maintained in the Signavio process modeling tool and linked to the ArchiMate model elements in Sparx EA. The network team conducted a bandwidth utilization study in February 2025, confirming that inter-datacenter traffic remains below sixty percent of the available WAN capacity under normal operating conditions.

6. Rollback Plan

In case of critical issues during migration, the rollback procedure is: (1) Stop all application connections to the new PostgreSQL instance, (2) Restore the Oracle database from the pre-migration backup on the SAN Storage Array, (3) Re-point application connection strings to Oracle, (4) Verify data integrity through automated reconciliation scripts. A full Oracle backup is also retained on the Backup Tape Library and the DR Recovery Node for 90 days post-migration.

Security review was completed in accordance with the CISO's security-by-design framework. The architectural principles that guide this work include separation of concerns, loose coupling between components, preference for open standards, and the ability to scale horizontally under peak load conditions such as end-of-year policy renewals. Dependencies on third-party services were documented and risk-assessed, with fallback mechanisms defined for each critical integration point. External consultants from Deloitte provided an independent assessment of the architecture in

late 2024. The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner. The enterprise technology radar, published semi-annually by the CTO's office, classifies technologies as Adopt, Trial, Assess, or Hold; the technologies used in this area are all currently in the Adopt or Trial rings. Capacity planning exercises revealed that the current infrastructure can sustain projected growth for the next twelve months, after which additional compute and storage resources will be required. The architecture team conducted a threat modeling exercise using the STRIDE framework, identifying potential spoofing, tampering, and information disclosure risks; mitigations for each threat were documented and tracked in the security risk register. The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints. The operational support model is defined in the Operational Runbook and includes on-call rotation and escalation procedures. The information security team requires that all new systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval. A chaos engineering program was introduced in Q1 2025, starting with controlled fault injection in the staging environment to validate the resilience of the system against network partitions and dependency failures. Cross-functional workshops held in January and March 2025 helped refine the scope and integration points. Documentation standards require that all architecture decisions are recorded as Architecture Decision Records using the ADR format proposed by Michael Nygard, stored alongside the codebase in the version control system. Automated deployment pipelines leverage the CI/CD infrastructure described in the platform engineering documentation. A comprehensive training program was developed for end users, including online modules, instructor-led sessions, and a sandbox environment for hands-on practice. The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards. Lessons learned from the pilot phase have been incorporated into the current design iteration. The transition plan was developed in alignment with the enterprise change management framework, incorporating staged rollouts, parallel-run periods, and defined go/no-go criteria at each milestone. The total cost of ownership analysis, prepared by the Finance Department in collaboration with IT, indicated a payback period of approximately eighteen months assuming current utilization rates. Change management activities include stakeholder communication, training sessions, and documentation updates. Cost allocation for shared infrastructure follows a chargeback model based on resource consumption metrics, with monthly reports sent to each business unit for transparency and accountability. Compliance with the General Data Protection Regulation requires that personally identifiable information is encrypted at rest and in transit, with access restricted to authorized roles as defined in the Identity and Access Management policy. The configuration management database maintains a record of all IT assets, including hardware serial numbers, software license keys, patch levels, and relationships between configuration items. From a historical perspective, ArchiSurance first recognized the need for modernization in this area following an internal efficiency review conducted in late 2022, which highlighted significant manual effort and process duplication across regional offices. A proof-of-concept was conducted over a four-week period, involving representatives from the Claims Department, IT Operations, and the Data Analytics Team to validate the technical feasibility. The vendor selection process followed the procurement guidelines established by the CFO's office. The team uses a definition of done that includes code review by at least two peers, passing CI pipeline, updated architecture diagrams, and entry in the change log maintained by the Release Manager. The Architecture Review Board convened a special session in February 2025 to evaluate the implications of the proposed changes on downstream systems and business processes. An accessibility audit was conducted by an external specialist in January 2025 to ensure that all user-facing components meet WCAG 2.1 Level AA standards, with remediation items tracked in the product backlog.

A formal lessons-learned workshop is held at the end of each quarter, with outcomes published to the internal wiki and action items assigned to specific team members with target completion dates. Budget allocation was approved as part of the annual IT investment portfolio review conducted in November 2024. An internal audit in March 2025 confirmed that the controls implemented for this area meet the requirements of the enterprise risk management framework. The vendor selection process followed the procurement guidelines established by the CFO's office. Security review was completed in accordance with the CISO's security-by-design framework. An accessibility audit was conducted by an external specialist in January 2025 to ensure that all user-facing components meet WCAG 2.1 Level AA standards, with remediation items tracked in the product backlog. The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards. Technical debt in this area was cataloged during a dedicated sprint in Q3 2024; the resulting backlog of forty-seven items has been prioritized by business impact and is being addressed alongside

feature development.

The Architecture Review Board convened a special session in February 2025 to evaluate the implications of the proposed changes on downstream systems and business processes. The enterprise data model, maintained by the Data Architecture team, defines canonical schemas for all shared data objects; integrating systems are expected to conform to these schemas or implement transformation layers that map proprietary formats to the canonical model. The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints. A dedicated Slack channel was established for real-time coordination between development, operations, and security teams during the rollout period. During the initial assessment phase, the team identified several areas where the existing architecture fell short of the target-state requirements defined in the enterprise roadmap. The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam. The vendor risk assessment framework evaluates suppliers across five dimensions: financial stability, technology roadmap alignment, support quality, security posture, and contractual flexibility. Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer.

Dependencies on third-party services were documented and risk-assessed, with fallback mechanisms defined for each critical integration point. Lessons learned from the pilot phase have been incorporated into the current design iteration. The enterprise technology radar, published semi-annually by the CTO's office, classifies technologies as Adopt, Trial, Assess, or Hold; the technologies used in this area are all currently in the Adopt or Trial rings. The data retention policy mandates that transactional records are kept for seven years in online storage and archived to cold storage thereafter, with the ability to retrieve archived records within twenty-four hours upon regulatory request. A chaos engineering program was introduced in Q1 2025, starting with controlled fault injection in the staging environment to validate the resilience of the system against network partitions and dependency failures. A comprehensive training program was developed for end users, including online modules, instructor-led sessions, and a sandbox environment for hands-on practice. The information security team requires that all new systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval. An accessibility audit was conducted by an external specialist in January 2025 to ensure that all user-facing components meet WCAG 2.1 Level AA standards, with remediation items tracked in the product backlog.

7. Testing Strategy

The testing strategy uses the CI/CD Build Server for automated regression testing. Test phases include: (1) Schema comparison, (2) Row-level data validation, (3) Application integration testing, (4) Performance benchmarking, (5) User acceptance testing.

Phase	Duration	Owner	Go/No-Go
Schema Validation	1 week	DBA Team	Automated
Data Reconciliati	2 weeks	Data Team	Automated + Manual
Integration Testi	2 weeks	App Teams	Manual
Performance Testi	1 week	Infra Team	Automated
UAT	2 weeks	Business Users	Manual sign-off

Stakeholders across multiple departments have contributed requirements for this workstream. The non-functional requirements specify a maximum end-to-end latency of five hundred milliseconds at the ninety-ninth percentile, measured under a sustained load of one thousand concurrent users. Post-implementation review sessions are scheduled for thirty and ninety days after go-live to capture lessons learned and identify optimization opportunities. Cost allocation for shared infrastructure follows a chargeback model based on resource consumption metrics, with monthly reports sent to each business unit for transparency and accountability. The architectural principles that guide this work include separation of concerns, loose coupling between components, preference for open standards, and the ability to scale horizontally under peak load conditions such as end-of-year policy renewals. Documentation standards require that all architecture decisions are recorded as Architecture Decision Records using the ADR format proposed by Michael

Nygard, stored alongside the codebase in the version control system. The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints. Monitoring and alerting are integrated with the central observability stack based on Prometheus and Grafana. The migration testing strategy validation has been a critical area of focus for ArchiSurance since the digital transformation initiative began in 2023.

Technical debt in this area was cataloged during a dedicated sprint in Q3 2024; the resulting backlog of forty-seven items has been prioritized by business impact and is being addressed alongside feature development. The design considers both current-state constraints and target-state aspirations as documented in the strategic roadmap. The incident management process defines four severity levels, with Severity 1 incidents requiring a response within fifteen minutes and a post-incident review within five business days. The information security team requires that all new systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval. The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline. Risk mitigation strategies include regular checkpoint reviews, automated regression testing, and staged rollout procedures. The non-functional requirements specify a maximum end-to-end latency of five hundred milliseconds at the ninety-ninth percentile, measured under a sustained load of one thousand concurrent users. The network team conducted a bandwidth utilization study in February 2025, confirming that inter-datacenter traffic remains below sixty percent of the available WAN capacity under normal operating conditions. The migration strategy follows a strangler-fig pattern, where new functionality is built in the target platform while legacy features are gradually redirected; this approach minimizes disruption to active users and allows incremental validation of the new implementation.

Capacity forecasting models project a twenty-five percent year-over-year increase in data volume, driven primarily by the digitization of paper-based processes and the expansion of the broker channel. External consultants from Deloitte provided an independent assessment of the architecture in late 2024. Scalability testing performed in the staging environment confirmed that the system handles a three-times-peak load without degradation, providing sufficient headroom for seasonal demand fluctuations. The architecture review confirmed that the proposed solution aligns with the TOGAF Architecture Development Method phases used by ArchiSurance, specifically phases B through D covering Business, Information Systems, and Technology Architecture. The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner. Performance benchmarks are captured in the enterprise monitoring framework and reported on a monthly basis to leadership. The business process documentation follows the BPMN 2.0 standard, with process models maintained in the Signavio process modeling tool and linked to the ArchiMate model elements in Sparx EA. Stakeholders across multiple departments have contributed requirements for this workstream. The enterprise technology radar, published semi-annually by the CTO's office, classifies technologies as Adopt, Trial, Assess, or Hold; the technologies used in this area are all currently in the Adopt or Trial rings.

8. Appendices

References and Related Documents

Doc ID	Title	Version
EA-OVR-001	Enterprise Architecture Overview	3.0
EA-APP-002	Application Portfolio & Inventory	4.2
EA-INFRA-003	Infrastructure Design Document	2.5
EA-BIZ-004	CRM Business Process Description	2.0
EA-CLM-005	Claims Processing Workflow	3.1
EA-SEC-006	Security Architecture	2.3
EA-INT-007	Integration Architecture	2.8
EA-DAT-008	Data Architecture & Governance	2.0
EA-CLD-009	Cloud Migration Assessment	1.5
EA-DR-010	Disaster Recovery & Business Contin	3.0
EA-NET-011	Network Architecture	2.2
EA-ROAD-012	Strategic IT Roadmap 2025-2027	2.0
EA-SVC-013	IT Service Catalog	3.5
EA-VND-014	Vendor & Technology Assessment	1.8
EA-OPS-015	Operational Runbook: Core Platform	4.1
EA-CMP-016	Compliance & Regulatory Requirement	2.5
EA-CHG-017	Change Management: ERP Migration	1.2

Distribution List

Name	Role	Responsibility
J. van den Berg	Enterprise Architect	Review & Approve
M. de Vries	Application Architect	Review & Contribute
P. Jansen	Integration Architect	Review & Contribute
K. Bakker	Infrastructure Architect	Review & Contribute
S. Visser	Business Analyst	Inform
T. Mulder	Process Architect	Inform
R. de Jong	CISO	Review & Approve
L. Hendriks	Data Architect	Inform
A. Willems	CTO	Final Approval
B. Smit	Head of Claims	Business Sign-off

Appendix: Meeting Minutes - Architecture Review Board

Date: March 15, 2025. Location: ArchiSurance HQ, Room 4.12. Attendees: J. van den Berg (chair), M. de Vries, P. Jansen, K. Bakker, S. Visser, T. Mulder (remote). Apologies: R. de Jong (on leave).

Agenda item 1: Review of open action items from previous meeting. Action 2024-47 (Jansen): Provide updated cost estimate for Oracle migration. Status: Completed. The revised estimate is EUR 1.4M including contingency. Action 2024-48 (de Vries): Schedule penetration test with external vendor. Status: In progress, vendor shortlist reduced to two candidates.

Agenda item 2: Discussion on coffee machine replacement in the 3rd floor kitchen. The current Jura E8 has been experiencing frequent bean grinder failures. Facilities will procure a replacement by end of month. Not related to architecture but raised due to impact on team productivity during long review sessions.

Agenda item 3: Parking lot discussion about whether to adopt Backstage as the internal developer portal. Deferred to next meeting pending PoC results. K. Bakker noted that the onboarding documentation is outdated and should be

refreshed regardless of the portal decision.

Next meeting: April 12, 2025 at 14:00 CET. Van den Berg to circulate agenda by April 5.

Detailed Revision Notes for Version 2.1

Section 2.3 was updated to reflect the new naming convention adopted by the Platform Engineering team. All references to 'microservice cluster' have been replaced with 'application workload group' per the Architecture Board decision of January 2025. Section 4.1 was expanded to include the results of the capacity planning exercise conducted in Q4 2024, which identified the need for additional compute resources in the Amsterdam primary data center.

Appendix B was added to include the RACI matrix requested by the PMO during the Q3 2024 quarterly review. The glossary was updated to include twelve new acronyms that were introduced since the previous version. Figure 3 was replaced with an updated diagram generated from the Sparx EA repository on March 28, 2025. The document classification was elevated from 'Internal' to 'Internal - Confidential' following the data classification review of February 2025.

Feedback from the peer review conducted by M. de Vries on March 10, 2025, addressed three items: (1) clarification of the boundary between the DMZ and the application tier, (2) addition of the backup retention policy reference, and (3) correction of the SLA target for the Claims Handling Service from 99.5% to 99.9% as agreed in the contract amendment of December 2024.

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